A Density-Deficit Obstruction for Near-Full Sidon Subsystems

Candidate partial result for arXiv:1504.05290

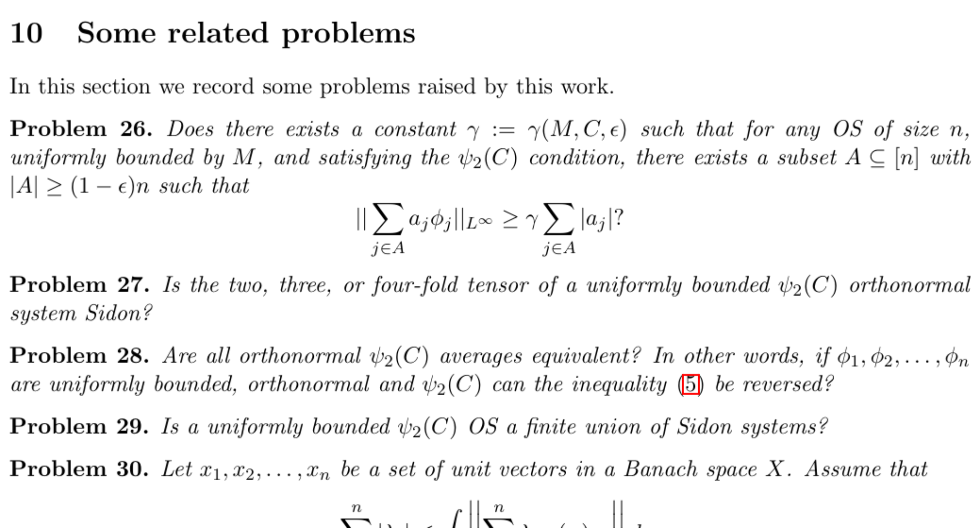
9 August 2026

Status: candidate partial result, likely valid. Problem 26 remains open for each fixed positive density deficit.

1 Source problem

Bourgain and Lewko ask the following in Problem 26 of [1] (arXiv PDF page 20).

> Does there exist a constant $\gamma = \gamma(M, C, \varepsilon)$ such that for any orthonormal system of size n, uniformly bounded by M and satisfying the $\psi_2(C)$ condition, there exists $A \subseteq [n]$ with $|A| \geq (1-\varepsilon)n$ such that
>
> $$\left\| \sum_{j \in A} a_j \phi_j \right\|_{L^\infty} \geq \gamma \sum_{j \in A} |a_j|?$$



10 Some related problems

In this section we record some problems raised by this work.

Problem 26. *Does there exists a constant $\gamma := \gamma(M, C, \epsilon)$ such that for any OS of size n, uniformly bounded by M, and satisfying the $\psi_2(C)$ condition, there exists a subset $A \subseteq [n]$ with $|A| \geq (1-\epsilon)n$ such that*

$$|| \sum_{j \in A} a_j \phi_j ||_{L^\infty} \geq \gamma \sum_{j \in A} |a_j|?$$

Problem 27. *Is the two, three, or four-fold tensor of a uniformly bounded $\psi_2(C)$ orthonormal system Sidon?*

Problem 28. *Are all orthonormal $\psi_2(C)$ averages equivalent? In other words, if $\phi_1, \phi_2, \ldots, \phi_n$ are uniformly bounded, orthonormal and $\psi_2(C)$ can the inequality (5) be reversed?*

Problem 29. *Is a uniformly bounded $\psi_2(C)$ OS a finite union of Sidon systems?*

Problem 30. *Let $x_1, x_2, \ldots, x_n$ be a set of unit vectors in a Banach space X. Assume that*

Figure 1: Problem 26 in arXiv:1504.05290, PDF page 20.

For a finite system $\mathcal{A} = (f_j)_{j\in A}$, write

$$\operatorname{sid}(\mathcal{A}) = \inf_{(a_j)\neq 0} \frac{\|\sum_{j\in A} a_j f_j\|_\infty}{\sum_{j\in A} |a_j|}$$

for its optimal Sidon lower constant.

2 Proof intuition

The Bourgain–Lewko counterexample has two coupled pieces. Its Rademacher piece would normally make equal coefficients large, but all functions share a factor which damps a Rademacher configuration according to the square of the total sign sum. If k coordinates have been deleted, those missing coordinates can neutralize at most k units of the retained sign sum. Past that threshold the common factor caps the retained contribution by $n/\sqrt{\log n}$. The Walsh–Rudin–Shapiro perturbation is also small: the full signed Walsh sum is $O(\sqrt{n})$, while deleting k summands costs at most k.

Equal coefficients therefore witness an upper Sidon constant of $k/(n-k) + o(1)$. Hence a density-$1-\varepsilon$ subsystem can have no universal constant larger than $\varepsilon/(1-\varepsilon)$, and deleting only $o(n)$ terms never suffices. This does not decide whether a positive constant of order at most ε always exists for fixed ε.

3 The quantitative obstruction

We use the real orthonormal systems from Section 9 of [1]. Fix large n, put $L = \log n$ and $\delta = \sqrt{L/n}$, let $r_1, \ldots, r_n$ be independent Rademacher functions, and choose Walsh functions W_i and signs σ_i so that

$$\left\| \sum_{i=1}^{n} \sigma_i W_i \right\|_\infty \leq 6\sqrt{n}. \tag{1}$$

The probability density used in [1] is

$$\Psi = \left(1 + \frac{L}{n}\right)^{-1} \left(1 + \frac{L}{n^2}\left(\sum_{i=1}^{n} r_i\right)^2\right).$$

For $1 \leq i \leq n$ their functions can consequently be written exactly as

$$\phi_i = D(r)(r_i - \delta\sigma_i W_i), \qquad D(r) = \left(1 + \frac{L}{n^2}\left(\sum_{i=1}^{n} r_i\right)^2\right)^{-1/2}. \tag{2}$$

Together with the additional function ϕ_0, these form an orthonormal system $\Phi^{(n)} = \{\phi_0, \phi_1, \ldots, \phi_n\}$, uniformly bounded by 7 and satisfying $\psi_2(C_0)$ for a universal C_0 [1].

Theorem 1. *Let $\mathcal{A} \subseteq \Phi^{(n)}$, put*

$$B = \{i \in \{1, \ldots, n\} : \phi_i \in \mathcal{A}\}, \qquad m = |B|, \qquad k = n - m,$$

and suppose $m > 0$. Then

$$\operatorname{sid}(\mathcal{A}) \leq \frac{k + n/\sqrt{\log n} + 6\sqrt{\log n} + k\sqrt{\log n/n}}{m}. \tag{3}$$

In particular, for every fixed $0 < \varepsilon < 1$ *and every choice* $\mathcal{A}_n \subseteq \Phi^{(n)}$ *satisfying* $|\mathcal{A}_n| \geq (1-\varepsilon)(n+1)$,

$$\limsup_{n\to\infty} \operatorname{sid}(\mathcal{A}_n) \leq \frac{\varepsilon}{1-\varepsilon}. \tag{4}$$

Proof. Assign coefficient 1 to the functions ϕ_i with $i \in B$ and coefficient 0 to ϕ_0 if it belongs to $\mathcal{A}$. This coefficient vector has ℓ_1 norm m. Write $K = \{1, \dots, n\} \setminus B$ and

$$S_B = \sum_{i\in B} r_i, \qquad S_K = \sum_{i\in K} r_i, \qquad P_B = \sum_{i\in B} \sigma_i W_i.$$

Since $|S_K| \leq k$,

$$|S_B + S_K| \geq (|S_B| - k)_+. \tag{5}$$

Moreover, (1) and $|W_i| = 1$ give

$$|P_B| \leq \left|\sum_{i=1}^{n} \sigma_i W_i\right| + \left|\sum_{i\in K} \sigma_i W_i\right| \leq 6\sqrt{n} + k. \tag{6}$$

Equations (2)–(6), with $x = |S_B|$, imply pointwise

$$\left|\sum_{i\in B} \phi_i\right| \leq \frac{x}{\sqrt{1 + (L/n^2)(x-k)_+^2}} + \delta(6\sqrt{n} + k).$$

The first term is at most $k + n/\sqrt{L}$. Indeed, it is at most k when $x \leq k$; when $x = k + y > k$, it is bounded by

$$\frac{k+y}{\sqrt{1+(L/n^2)y^2}} \leq k + \frac{y}{\sqrt{1+(L/n^2)y^2}} \leq k + \frac{n}{\sqrt{L}}.$$

Taking the supremum and using $\delta(6\sqrt{n} + k) = 6\sqrt{L} + k\sqrt{L/n}$ proves (3).

If $|\mathcal{A}_n| \geq (1-\varepsilon)(n+1)$, then

$$m \geq |\mathcal{A}_n| - 1 \geq (1-\varepsilon)n - \varepsilon, \qquad k \leq \varepsilon n + \varepsilon.$$

Divide (3) by m and let $n \to \infty$. Every term except k/m tends to zero, while $\limsup k/m \leq \varepsilon/(1-\varepsilon)$. This proves (4). ☐

Corollary 2. *There are universally bounded orthonormal* $\psi_2(C_0)$ *systems for which no subsystem obtained by deleting* $o(n)$ *functions has a Sidon lower constant bounded away from zero. Consequently, if Problem 26 has an affirmative answer, its best possible universal dependence for* $M = 7, C = C_0$ *satisfies*

$$\gamma(7, C_0, \varepsilon) leq \frac{\varepsilon}{1-\varepsilon} \qquad (0 < \varepsilon < 1).$$

Proof. For the first assertion apply Theorem 1 with $k = o(n)$. For the second, a constant asserted by Problem 26 must work for every member of the Bourgain–Lewko family. The bound (4) applies to every subsystem of the required size, so no larger universal constant is possible. ☐

4 Limitations and novelty check

The theorem is not a counterexample to Problem 26 for fixed $\varepsilon > 0$: the right side of (4) is then positive, and the problem explicitly allows γ to depend on ε. What is settled is the stronger density-one version with $o(n)$ deletions and an explicit linear upper restriction on any possible fixed-ε answer.

A bounded arXiv and local full-source search used arXiv:1504.05290, papers citing it, the exact phrase of Problem 26, and the terms "density-one Sidon subsystem", "Sidon lower constant", and "uniformly bounded ψ_2 orthonormal system". Pisier's later work resolves the two-fold tensor question but not Problem 26. No source stating Theorem 1 or its corollary was located. This supports, but does not certify, novelty.

5 Verification and human-review focus

The proof is elementary once the source construction is written as (2). The common-factor identity, the two triangle inequalities (5)–(6), and all limiting quantifiers were checked separately in the accompanying verification note. No numerical computation is used.

Human review should focus on the exact rewriting of the density Ψ and on the use of zero coefficient for ϕ_0 when the tested subsystem contains that function.

References

[1] J. Bourgain and M. Lewko, *Sidonicity and variants of Kaczmarz's problem*, Ann. Inst. Fourier (Grenoble) 67 (2017), no. 3, 1321–1352; arXiv:1504.05290.